

DevOps & CI/CD

Continuous integration, continuous delivery/deployment, pipelines, Quality Gates, Docker and environments.

What DevOps is

DevOps combines cultural and technical practices to bring development and operations closer, reduce delivery friction and increase feedback. QA contributes continuous quality throughout the flow.

CI — Continuous Integration

CI integrates changes frequently and runs automated checks such as builds, tests and static analysis. Fast failures help teams fix problems before they accumulate.

CD — Delivery and Deployment

Continuous Delivery keeps software releasable; Continuous Deployment also automates release to production when criteria are met. The terms should be used with context.

Pipeline

A pipeline may run checkout, dependency installation, build, unit/API/UI/security tests, artifact generation and deployment. Quality Gates can stop the flow when criteria fail.

GitHub Actions

GitHub Actions defines workflows versioned with the repository. QA can run Pytest/Playwright, publish reports and store evidence.

Environments and configuration

DEV, QA, SIT, UAT, Staging and Production may have different purposes. Environment variables and secrets separate configuration from code and avoid hardcoded credentials.

Docker and reproducibility

Containers help run applications and dependencies consistently. QA can use them to reproduce services, databases and tools in pipelines.

Shift Left, Shift Right and Continuous Testing

Shift Left brings feedback earlier; Shift Right observes quality in production; Continuous Testing integrates appropriate checks into delivery. None means running every test at every stage.

QA tip

CI/CD is not just “deploying”: it is a feedback and delivery flow that can include multiple quality controls.